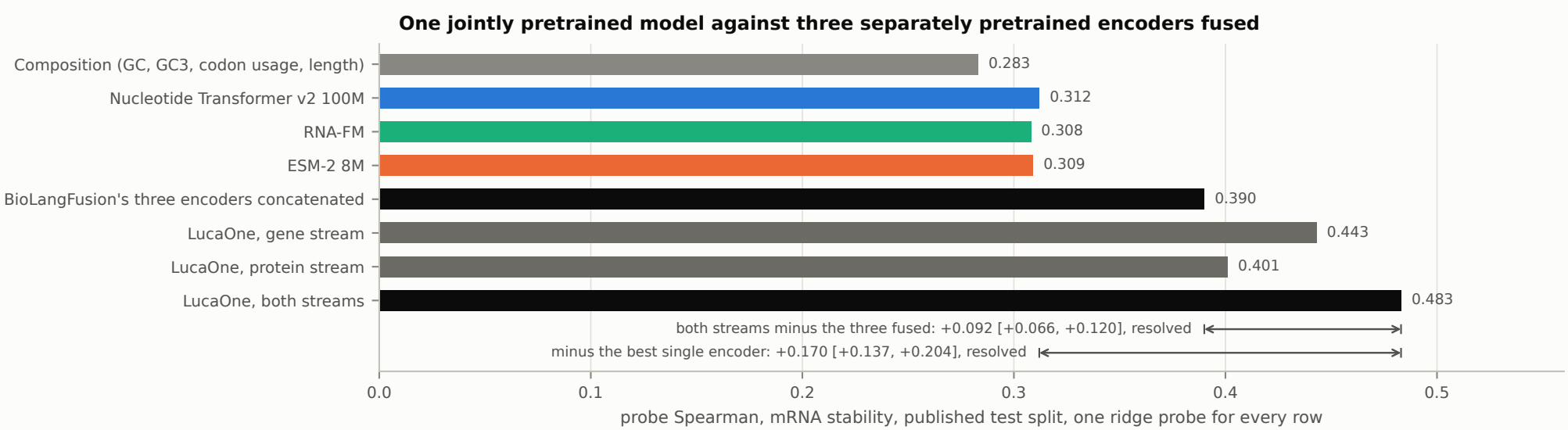
One jointly pretrained model against three separately pretrained encoders fused
Composition (GC, GC3, codon usage, length)
Nucleotide Transformer v2 100M
RNA-FM
ESM-2 8M
BioLangFusion's three encoders concatenated
LucaOne, gene stream
LucaOne, protein stream
LucaOne, both streams
0.283
0.312
0.308
0.309
0.390
0.443
0.401
0.483
both streams minus the three fused: +0.092 [+0.066, +0.120], resolved
minus the best single encoder: +0.170 [+0.137, +0.204], resolved
0.0
0.1
0.2
0.3
0.4
0.5
probe Spearman, mRNA stability, published test split, one ridge probe for every row